

Name: _____ Date: _____

Period: _____

Parametrically Defined Circles and Lines

1. A circle is centered at $(3, -1)$ with radius 4. Write parametric equations for *counterclockwise* motion that starts at $(7, -1)$.

$x(t) = \underline{\hspace{2cm}} \qquad y(t) = \underline{\hspace{2cm}}$

Domain: _____

2. Write a parametrization for the line segment from $(0, 6)$ to $(4, 0)$.

$x(t) = \underline{\hspace{2cm}} \qquad y(t) = \underline{\hspace{2cm}}$

Domain: _____

At what value of t does the segment cross the x -axis (where $y = 0$)? Show your algebra.
